Tutorial 1

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Suppose two points (x_0, y_0) and (x_1, y_1) are on a straight line with $y_1 \neq y_0$. Two formulas are available to find the x-intercept of the line:

$$x = \frac{x_0 y_1 - x_1 y_0}{y_1 - y_0} \quad \text{and} \quad x = x_0 - \frac{(x_1 - x_0) y_0}{y_1 - y_0}.$$

a. Show that both formulas are algebraically correct.

b. Use the data $(x_0, y_0) = (1.31, 3.24)$ and $(x_1, y_1) = (1.93, 4.76)$ and three-digit rounding arithmetic to compute the x-intercept both ways. Which method is better and why?

a) $$x = \frac{x_0 y_1 - x_1 y_0}{y_1 - y_0} = \frac{x_0 y_1 + x_0 y_0 - x_0 y_0 - x_1 y_0}{y_1 - y_0}$$

$$= \frac{x_0 (y_1 - y_0) + y_0 (x_0 - x_1)}{y_1 - y_0} = x_0 + \frac{y_0 (x_0 - x_1)}{(y_1 - y_0)}$$

b) 1) $$x = \frac{(1.31)(4.76) - (3.24)(1.93)}{4.76 - 3.24}$$

$$= \frac{6.24 - 6.25}{1.52} = -\frac{0.0100}{1.52}$$

$$= -0.00658$$

2) $$x = 1.31 + \frac{3.24(1.31 - 1.93)}{(4.76 - 3.24)}$$

$$= 1.31 + \frac{3.24(-0.620)}{1.52}$$

$$= 1.31 - \frac{2.01}{1.52} = 1.31 - 1.32$$

$$= 0.0100$$

Actual value $= -0.0115$

Relative error with (1) 42.8%

" " with (2) 13.0%

Formula (2) is better since steps are less thus less rounding error.

The polynomial $P_2(x) = 1 - \frac{1}{2}x^2$ is to be used to approximate $f(x) = \cos x$ in $[-\frac{1}{2}, \frac{1}{2}]$. Find a bound for the maximum error.

$$R_3(x) = \frac{f^{(4)}(\xi)}{24} x^4 = \frac{\cos\xi}{24} x^4$$

$$|R_3(x)| \leq \frac{|\cos\xi|}{24} (0.5)^4 \leq \frac{(0.5)^4}{24} = 2.604 \times 10^{-3}$$

Let $f(x) = e^x$ and $x_0 = 0$. Find the nth Taylor polynomial $P_n(x)$ for $f(x)$ about x_0. Find a value of n necessary for $P_n(x)$ to approximate $f(x)$ to within 10^{-6} on $[0, 0.5]$.

$$f^{(n)}(x) = e^x \quad \forall n, \quad f^{(n)}(0) = 1$$

$$P_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}$$

$$R_n(x) = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} x^{n+1} \right| \leq \frac{(0.5)^{n+1}}{(n+1)!} \leq 10^{-6}$$

by suitable substitution $n \geq 7$.

Use a Taylor polynomial about $\pi/4$ to approximate $\cos 42^\circ$ to an accuracy of 10^{-6}.

So, first we need to estimate n such that the error would be less that 10^{-6}.

$$|R_n(x)| = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - \tfrac{\pi}{4})^{n+1} \right|$$

Since $f(x) = \cos x$ derivatives of $f(x)$ are either $\sin/\cos$
$|f^{(n)}(x)| \leq 1 \quad \forall n.$

$$\Rightarrow |R_n(x)| \leq \frac{(x - \pi/4)^{n+1}}{(n+1)!} \quad \text{Since } 42^\circ = 7\pi/30$$

$$\leq \frac{(\frac{7\pi}{30} - \pi/4)^{n+1}}{(n+1)!} \leq 10^{-6}$$

by suitable substitution we can see that $n = 3$

$$\frac{(\frac{7\pi}{30} - \pi/4)^4}{4!} \simeq 3.1 \times 10^{-7}$$

Thus, it is sufficient to calculate $P_3(x)$

$$f(\pi/4) = \frac{\sqrt{2}}{2}, \quad f'(\pi/4) = -\frac{\sqrt{2}}{2}, \quad f''(\pi/4) = -\frac{\sqrt{2}}{2}$$
$$f'''(\pi/4) = \frac{\sqrt{2}}{2}$$

$$P_3(x) = \frac{\sqrt{2}}{2}\left(1 - (x-\tfrac{\pi}{4}) - \frac{1}{2}(x-\tfrac{\pi}{4})^2 + \frac{1}{6}(x-\tfrac{\pi}{4})^3\right)$$

$$P_3\left(\frac{7\pi}{30}\right) \simeq 0.7431446$$

$$\left|f(\pi/4) - P_3\left(\frac{7\pi}{30}\right)\right| = 2.25 \times 10^{-7}.$$

Another way is done by constructing a table.

	$\cos(\frac{7\pi}{30})$	ε_t	ε_a
$P_0(x) = \frac{\sqrt{2}}{2}$	0.7071068	4.8×10^{-2}	—
$P_1(x) = \frac{\sqrt{2}}{2}\left(1 - (x-\pi/4)\right)$	0.7441308	1.33×10^{-3}	5.0×10^{-2}
$P_2(x) = \frac{\sqrt{2}}{2}\left(1 - (x-\pi/4) - \frac{1}{2}(x-\pi/4)^2\right)$	0.7431615	2.2×10^{-5}	1.3×10^{-3}
$P_3(x) = \frac{\sqrt{2}}{2}\left(1 - (x-\pi/4) - \frac{1}{2}(x-\pi/4)^2 + \frac{1}{6}(x-\pi/4)^3\right)$	0.7431446	2.3×10^{-7}	2.27×10^{-5}

Find the third Taylor polynomial $P_3(x)$ for the function $f(x) = (x-1)\ln x$ about $x_0 = 1$.

a. Use $P_3(0.5)$ to approximate $f(0.5)$. Find an upper bound for error $|f(0.5) - P_3(0.5)|$ using the error formula, and compare it to the actual error.

b. Find a bound for the error $|f(x) - P_3(x)|$ in using $P_3(x)$ to approximate $f(x)$ on the interval $[0.5, 1.5]$.

c. Approximate $\int_{0.5}^{1.5} f(x)\,dx$ using $\int_{0.5}^{1.5} P_3(x)\,dx$.

d. Find an upper bound for the error in (c) using $\int_{0.5}^{1.5} |R_3(x)\,dx|$, and compare the bound to the actual error.

$f(1) = 0 \qquad f'(1) = \ln x + \frac{(x-1)}{x}\Big|_1 = 0$

$f''(1) = \frac{1}{x} + \frac{1}{x^2}\Big|_1 = 2 \qquad f'''(1) = -\frac{1}{x^2} - \frac{2}{x^3}\Big|_{x=1} = -3$

$P_3(x) \simeq (x-1)^2 - \frac{1}{2}(x-1)^3$

upper bound $|f(x) - P_3(x)| \le |R_3(x)|$

$= \left| \frac{f^{(4)}(\xi)}{4!}(x-1)^4 \right| = \frac{1}{24}\left| \frac{2}{\xi^3} + \frac{6}{\xi^4} \right| |(x-1)^4|$

$\xi \in [0.5, 1.5]$ least value 0.5

$\frac{2}{|\xi|^3}\left|1 + \frac{3}{\xi}\right| \le \frac{2}{|\xi|^3}\left(1 + \frac{3}{|\xi|}\right) \le 16(1+6) = 112$

$|f(0.5) - P_3(0.5)| \le \frac{112}{24}(0.5-1)^4 = 0.2916$

actual error $f(0.5) = 0.346574$

$P_3(0.5) = 0.312500$

$|f(0.5) - P_3(0.5)| \simeq 0.034074.$

c) $\int_{0.5}^{1.5} f(x)\,dx \simeq \int_{0.5}^{1.5} P_3(x)\,dx \simeq \int_{0.5}^{1.5} \left[(x-1)^2 - \frac{1}{2}(x-1)^3\right] dx$

$= \frac{(x-1)^3}{3} - \frac{1}{8}(x-1)^4\Big|_{0.5}^{1.5} = 0.083333$

$$= \frac{(x-1)^3}{3} - \frac{1}{8}(x-1)^4 \Big|_{0.5} = 0.083333$$

Actual $\int_{0.5}^{1.5} (x-1)\ln x \, dx$

$u = \ln x \qquad v = \frac{(x-1)^2}{2}$

$du = \frac{1}{x}dx \qquad dv = (x-1)dx$

$$\ln x \, \frac{(x-1)^2}{2} \Big|_{0.5}^{1.5} - \frac{1}{2}\int_{0.5}^{1.5} \frac{x^2-2x+1}{x}\,dx$$

$$\int_{0.5}^{1.5} \left(x - 2 + \frac{1}{x}\right)dx = \frac{x^2}{2} - 2x + \ln(x) \Big|_{0.5}^{1.5}$$

$\simeq 0.088020$

d) error bound $\int_{0.5}^{1.5} |R_3(x)|\,dx \simeq \frac{112}{24}\int_{0.5}^{1.5} (x-1)^4\,dx$

$$= \frac{112}{24}\,\frac{(x-1)^5}{5}\Big|_{0.5}^{1.5} \simeq 0.058333$$

Actual error $|0.083333 - 0.088020| = 0.004687.$

Find the third Taylor polynomial $P_3(x)$ for the function $f(x) = \sqrt{x+1}$ about $x_0 = 0$. Approximate $\sqrt{0.5}$, $\sqrt{0.75}$, $\sqrt{1.25}$, and $\sqrt{1.5}$ using $P_3(x)$, and find the actual errors.

$$f(x) = \sum_{n=0}^{3} \frac{f^{(n)}(0)}{n!}x^n + R_3(x)$$

$f(0) = 1, \quad f'(0) = \frac{1}{2\sqrt{x+1}} = \frac{1}{2f(x)}\Big|_{x=0} = \frac{1}{2}$

$f''(0) = \frac{1}{2}\frac{(-1)}{f(x)^2}f'(x) = -\frac{1}{2}\left(\frac{1}{2f(x)}\right)\frac{1}{f(x)^2} = -\frac{1}{4f(x)^3}\Big|_{x=0} = -\frac{1}{4}$

$f'''(0) = \left(-\frac{1}{4}\right)\left(-3\frac{1}{f(x)^4}\right)\left(\frac{1}{2f(x)}\right)\Big|_{x=0} = \frac{3}{8}$

$$f(x) = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + R_3(x)$$

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$$\sqrt{0.5+1} \simeq 1 + \frac{1}{2}\left(\frac{1}{2}\right) - \frac{1}{8}\left(\frac{1}{2}\right)^2 + \frac{1}{16}\left(\frac{1}{2}\right)^3 \simeq 1.2265625$$

$$R_3(0.5) = \left| \sqrt{0.5+1} - P_3(0.5) \right| = 0.0018176$$

and so, on.

In this application, we were given $P = 1.00$ atm, $V = 0.100$ m^3, $N = 0.00420$ mol, and $R = 0.08206$. Solving for T in the ideal gas law gives

$$T = \frac{PV}{NR} = \frac{(1.00)(0.100)}{(0.00420)(0.08206)} = 290.15 \text{ K} = 17^\circ\text{C}.$$

In the laboratory, it was found that T was 15°C under these conditions, and when the pressure was doubled and the volume halved, T was 19°C. Assume that the data are rounded values accurate to the places given, and show that both laboratory figures are within the bounds of accuracy for the ideal gas law.

$P = 1 \pm 0.005$, $V = 0.1 \pm 0.0005$

$N = 4.2 \times 10^{-3} \pm 5 \times 10^{-6}$ $R = 0.08206 \pm 0.0000005$

So, we want to see how error propagates within this formula

$$\Delta T = \left|\frac{\partial T}{\partial P}\right| \Delta P + \left|\frac{\partial T}{\partial V}\right| \Delta V + \left|\frac{\partial T}{\partial N}\right| \Delta N + \left|\frac{\partial T}{\partial R}\right| \Delta R$$

$$= \left|\frac{V}{NR}\right| \Delta P + \left|\frac{P}{NR}\right| \Delta V + \left|\frac{PV}{N^2 R}\right| \Delta N + \left|\frac{PV}{NR^2}\right| \Delta R$$

$= 3.264570$ ← actual error

$13.735 \leq T \leq 20.27$

relative error $\frac{\Delta T}{T} = 0.1920$

So, $T = 15$ ← is within acceptable range of error

Since P is doubled , V halved

$P = 2 \pm 0.01$ $V = 0.05 \pm 0.00025$

$N = 4.2 \times 10^{-3} \pm 5 \times 10^{-6}$ $R = 0.08206 \pm 0.0000005$

$$\Delta T = \left|\frac{V}{NR}\right|\Delta P + \left|\frac{P}{NR}\right|\Delta V + \left|\frac{PV}{N^2R}\right|\Delta N + \left|\frac{PV}{NR^2}\right|\Delta R$$

$$= 2.9193$$

$$14.0807 \leq T \leq 19.9193$$

measured $T = 19$ so still in acceptable range

Another way to do it by considering maximum possible error bounds on T as we did in the lecture

Find $O(\Delta x^2)$ forward difference of $f''(x)$

we want forward difference thus, we can only consider $f(x)$ $f(x+\Delta x)$, $f(x+2\Delta x)$ and so on. Note the order Δx^2 since $f''(x)$ has a factor of $(\Delta x)^2$, we should at least consider taylor polynomial up to 3th order,

$$f(x+\Delta x) = f(x) + \Delta x\, f'(x) + \frac{\Delta x^2}{2!} f''(x) + \frac{\Delta x^3}{3!} f'''(x) + \frac{\Delta x^4}{4!} f^{(4)}(\xi_1)$$

$$f(x+2\Delta x) = f(x) + 2\Delta x\, f'(x) + \frac{4\Delta x^2}{2!} f''(x) + \frac{8\Delta x^3}{3!} f'''(x) + \frac{16\Delta x^4}{4!} f^{(4)}(\xi_2)$$

we can remove the first derivative by multiplying 1st equation by 2 and substraction however, we would still have 3rd order derivative. To remove it we have to consider another point.

$$f(x+3\Delta x) = f(x) + 3\Delta x f'(x) + \frac{9\Delta x^2}{2!} + \frac{27\Delta x^2}{3!} f''(x) + \frac{81\Delta x^4}{4!} f^{(4)}(\xi_3)$$

So, we want to multiply by 3 coeffts A, B, C

such that,

$$(A + 2B + 3C) = 0 \quad , \quad (A + 8B + 27C) = 0$$

$$A = -2B - 3C \Rightarrow 6B + 24C = 0 \Rightarrow B = -4C$$

$$\Rightarrow C = 1 \ , \ B = -4 \ , \ A = 5$$

$$5f(x+\Delta x) - 4f(x+2\Delta x) + f(x+3\Delta x)$$

$$= 2f(x) + \frac{\Delta x^2}{2}(-2) f''(x) + \frac{\Delta x^4}{4!}(\ )$$

$$f''(x) = \frac{2f(x) - 5f(x+\Delta x) + 4f(x+2\Delta x) - f(x+3\Delta x)}{\Delta x^2} + O(\Delta x^2)$$

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Find $O(h^4)$ central difference for $f'(x)$

by analogy 2-points in central difference produce h^2 Thus, 4 points would produce h^4. Thus, we need to consider $f(x+\Delta x)$, $f(x-\Delta x)$ $f(x+2\Delta x)$ & $f(x-2\Delta x)$.

① $f(x+\Delta x) = f(x) + \Delta x f'(x) + \frac{\Delta x^2}{2} f''(x) + \frac{\Delta x^3}{3!} f^{(3)}(x) + \frac{\Delta x^4}{4!} f^{(4)}(x) + \frac{\Delta x^5}{5!} f^{(5)}(\xi_1)$

② $f(x-\Delta x) = f(x) - \Delta x f'(x) + \frac{\Delta x^2}{2} f'' - \frac{\Delta x^3}{3!} f^{(3)} + \frac{\Delta x^4}{4!} f^{(4)} - \frac{\Delta x^5}{5!} f^{(5)}(\xi_2)$

③ $f(x+2\Delta x) = f(x) + 2\Delta x f'(x) + \frac{4\Delta x^2}{2} f'' + \frac{8\Delta x^3}{3!} + \frac{16\Delta x^4}{4!} f^{(4)} + \frac{32\Delta x^5}{5!} f^{(5)}(\xi_3)$

④ $f(x-2\Delta x) = f(x) - 2\Delta x f'(x) + \frac{4\Delta x^2}{2} f'' - \frac{8\Delta x^3}{3!} + \frac{16\Delta x^4}{4!} f^{(4)} - \frac{32\Delta x^5}{5!} f^{(5)}(\xi_4)$

Notice we want $O(\Delta x^4)$ Since $f'(x)$ is multiplied by Δx we have to go to fifth order. This time the equations are easier since they are symmetric.

We can get rid of ... with ... plus ...

they are symmetric.

we can get rid of 2nd and 4th derivative when directly making ① − ② & ③ − ④

$$\Rightarrow ⑥\quad f(x+\Delta x) - f(x-\Delta x) = 2\Delta x f'(x) + \frac{\Delta x^3}{3} f^{(3)}(x) + O(\Delta x^5)$$

$$⑦\quad f(x+2\Delta x) - f(x-2\Delta x) = 4\Delta x f'(x) + \frac{8\Delta x^3}{3} f^{(3)}(x) + O(\Delta x^5)$$

Now, it's clear how to get rid of 3rd derivative multiply ⑥ × 8 − ⑦

$$\Rightarrow 8f(x+\Delta x) - 8f(x-\Delta x) - f(x+2\Delta x) + f(x+2\Delta x)$$

$$= 12\Delta x f'(x) + O(\Delta x^5)$$

$$\Rightarrow f'(x) = \frac{-f(x+2\Delta x) + 8f(x+\Delta x) - 8f(x-\Delta x) + f(x-2\Delta x}{12\Delta x} + O(\Delta x^4).$$